

CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$.

Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$:

String	Accepted by $\mathcal{A}_1$?	Accepted by $\mathcal{A}_2$?
aaa	No	Yes
aab	Yes	Yes
aba	Yes	No
abb	Yes	No
baa	No	Yes
bab	No	No
bba	No	No
bbb	No	No

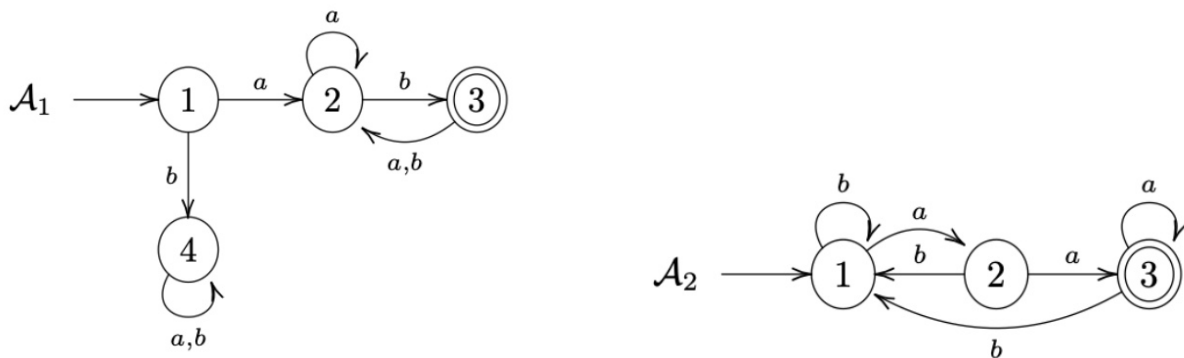


Figure 1: DFAs $\mathcal{A}_1$ (left) and $\mathcal{A}_2$ (right).

Exercise 1.2

describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Answer:

$L(\mathcal{A}_1)$: The language accepted by $\mathcal{A}_1$ consists of all strings over $\{a, b\}$ that start with a and contain at least one b . Starting with b leads to the trap state 4 and is rejected. $L(\mathcal{A}_2)$: The language accepted by $\mathcal{A}_2$ consists of all strings over $\{a, b\}$ that contain the substring aa . Reading two consecutive a 's leads to the accepting state 3.

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba
3. All the words that contain an odd number of a 's and an odd number of b 's
4. All the words that contain an even number of a 's and an odd number of b 's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answer: Each transition leads to a finalized state.

Thought process. I decided to map out a truth table and filled in all of the conditions and found that the outputs were satisfied at every condition, leading me to the conclusion that every transition was a satisfactory state condition.

Question

How might you combine the languages $L(\mathcal{A}_1)$ and $L(\mathcal{A}_2)$ from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

2.2 Week 2: Operations on Automata

Product automata

Exercise 1 (Product automata)

Consider the following automata $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$.

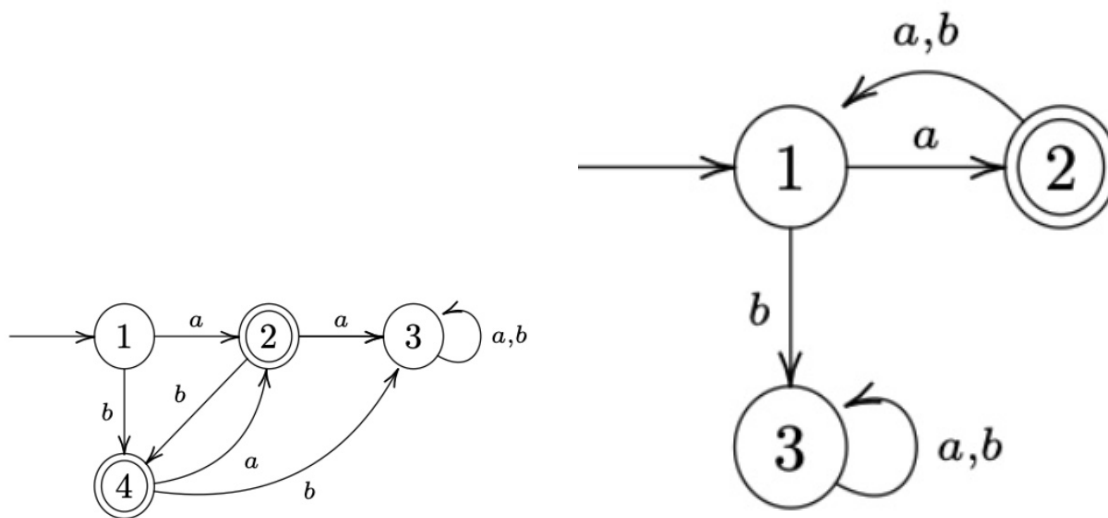


Figure 2: $\mathcal{A}^{(1)}$ (left) and $\mathcal{A}^{(2)}$ (right).

1. Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.

Answer:

- $L(\mathcal{A}^{(1)})$: Words include $a, b, ba, bab, baba, ab, aba, abab$, etc. I can see that each character must alternate based on its successive character.
- $L(\mathcal{A}^{(2)})$: Words include a, axa (where x doesn't matter). I see the word must start with a , and must add to its word length in increments of 2, every other character being a .

2. Construct the intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ by drawing its transition diagram (omit unreachable states).

Answer: Below is the intersection automaton diagram.

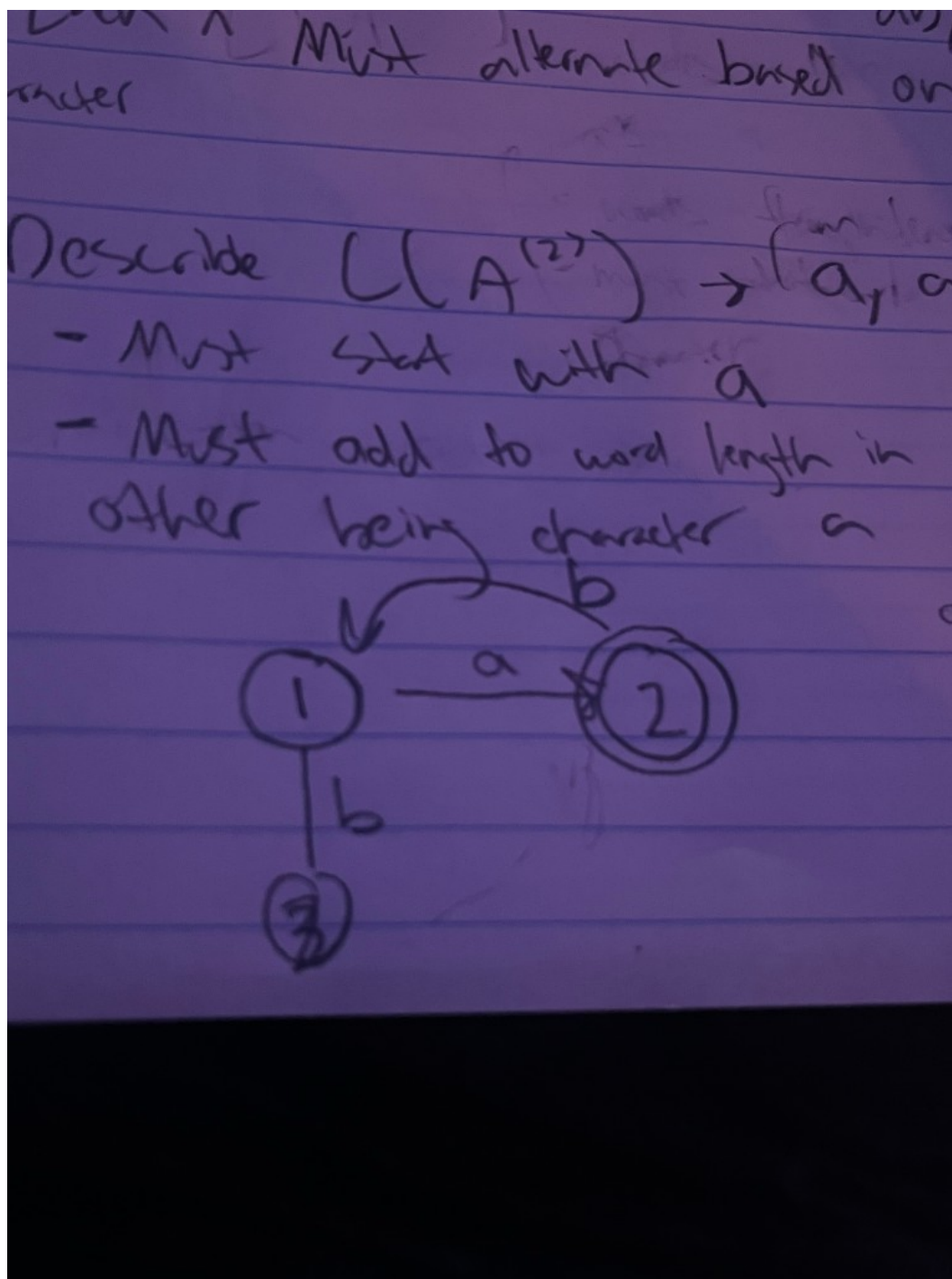


Figure 3: Intersection automaton $\mathcal{A}$ (hand-drawn diagram).

3. **Argue that** $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$.

Answer: Since $L(\mathcal{A}^{(1)})$ includes words such as a , b , ba , bab , $baba$ and ab , aba , $abab$, and $L(\mathcal{A}^{(2)})$ includes words that (1) must start with a —so we can disregard the words that start with b in $L(\mathcal{A}^{(1)})$,

leaving us with a, ab, aba , etc.—and (2) the requirement for $L(\mathcal{A}^{(2)})$ is to be in increments of two, we are left with patterns: $a, aba, ababa$, etc. Thus we obtain the automaton of both intersection.

4. **How can you change $\mathcal{A}$ to get an automaton $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$?**

Answer: We can allow for both word combinations to be accepting and add more state transitions.

Exercise 2 (More automata)

Consider the following automata $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$.

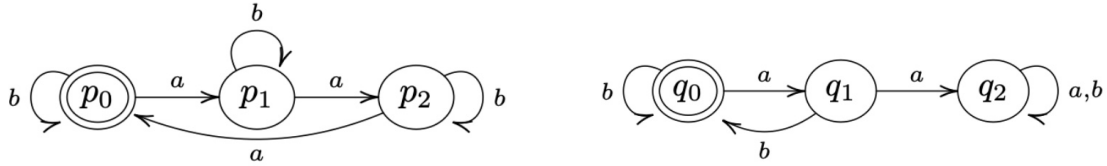


Figure 4: $\mathcal{B}^{(1)}$ (left) and $\mathcal{B}^{(2)}$ (right).

(Recall: the complement of a language $L \subseteq \Sigma^*$ is $\bar{L} := \Sigma^* \setminus L$.)

1. **Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.**

Answer:

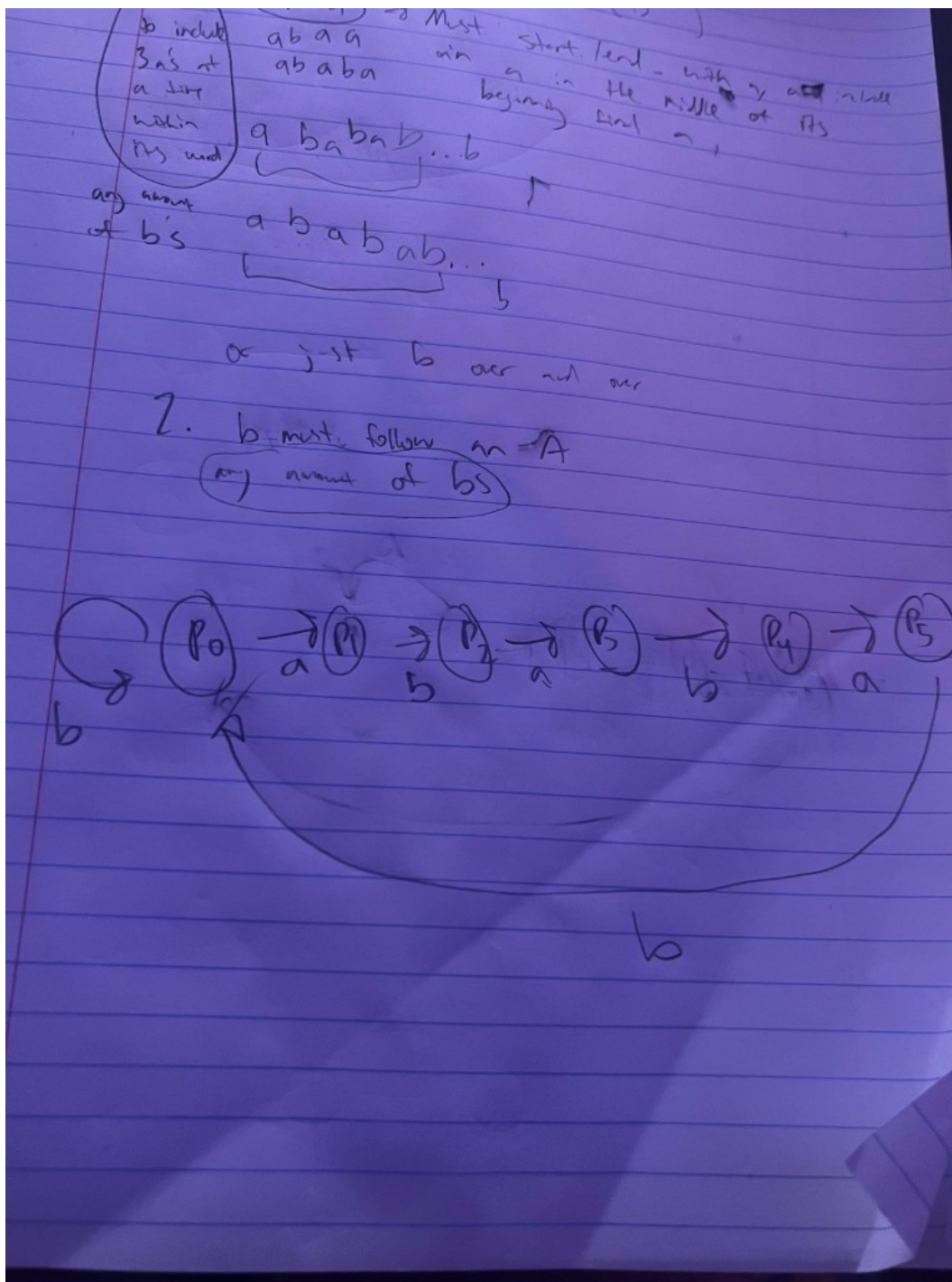


Figure 5: Handwritten notes: language rules for $B^{(1)}$ (e.g. include 3 a's at a time, any amount of b's) and $B^{(2)}$ (e.g. b must follow an a, any amount of b's), and DFA diagram for language 2.

2. Construct the intersection automaton B for $B^{(1)}$ and $B^{(2)}$ (omit unreachable states).

Answer: The product automaton has states (p_i, q_j) for $i, j \in \{0, 1, 2\}$, initial state (p_0, q_0) , and

accepting set $F = \{p_0\} \times \{q_0\} = \{(p_0, q_0)\}$. The transition rule is $\delta((p_i, q_j), \sigma) = (\delta^{(1)}(p_i, \sigma), \delta^{(2)}(q_j, \sigma))$. All 9 states are reachable from (p_0, q_0) . A word is accepted by $\mathcal{B}$ iff it is accepted by both $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$, so $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$, i.e. words where $\#_a(w) \equiv 0 \pmod{3}$, $\#_a(w)$ is even (hence $\#_a(w) \equiv 0 \pmod{6}$), and w has no substring aa . One can draw the 9-state diagram by applying the product transitions; the only accepting state is (p_0, q_0) .

3. **Argue that $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$.**

Answer: Both automata have the rule that any amount of b 's is allowed. But we must combine the rules regarding a 's. $L(\mathcal{B}^{(1)})$ requires 3 a 's to be included at a time per word, while $\mathcal{B}^{(2)}$ requires the rule that a must be followed by a b . This gives us the conclusion that words that do include an a must come in pairs of 3 a 's with b following the last a .

4. **How can you change $\mathcal{B}$ to get an automaton $\mathcal{B}'$ with $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$?**

Answer: We must flip the accepting states of automaton 1 to include: no amount of b 's AND words that include 3 a 's at a time only.

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, [Title](#), Publisher, Year.